

ALI NIKKHAH

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SUMMARY

Data Engineer and Data Scientist with end-to-end ownership of production data infrastructure: lakes, warehouses, real-time streaming pipelines, and ML-powered analytics. At Turquoise Digital, architected the central enterprise data lake for a BNPL and stock-trading platform (Airflow DAGs, PeerDB CDC replication, Trino/Parquet), built a real-time A/B testing engine, and developed an AI-driven BNPL credit risk model. At Digikala, designed distributed Spark feature-extraction pipelines and optimized vector indexing at scale across Elasticsearch, FAISS, and OpenSearch.

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer

Sep 2025 – Mar 2026

- Architected the central enterprise data lake and OLAP warehouse for a high-volume BNPL and stock-trading platform: routed transactional and user-interaction streams from PostgreSQL into **Trino** via **PeerDB** Change Data Capture (CDC) replication (**¡60s CDC latency, 5TB/day**); designed multi-stage **Airflow** DAGs for transformation, cold-data archiving to Parquet/ORC on object storage, and cross-system synchronization.
- Built a real-time A/B statistical inference engine running parallel hypothesis tests on application features, monitoring user-activity telemetry and automatically flagging significance shifts under volatile market conditions (**50+ concurrent experiments, mSPRT sequential testing**).
- Refactored legacy ETL scripts into modular **PySpark** pipelines improving memory efficiency (**60% memory reduction**); configured **Prometheus**, **Grafana**, and **Great Expectations** for end-to-end pipeline observability and automated data quality gating at ingestion.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Data Scientist

Aug 2025 – Sep 2025

- Developed predictive ML models analyzing BNPL user repayment histories and behavioral telemetry to compute dynamic credit limits; automated fund allocation and containment decisions with full audit logging; aligned historical price time-series with news-sentiment embeddings using time-series transformers for portfolio valuation forecasting (**default rate -22%, AUC 0.87**).

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Designed distributed **Apache Spark** feature-extraction jobs to pre-compute, partition, and serialize massive product-text datasets prior to batch embedding generation, parallelizing tokenization across the cluster to cut pipeline wall-time (**4x faster, 10M+ docs/day**).
- Optimized hybrid dense/sparse vector indexing across **Elasticsearch**, **OpenSearch**, and **FAISS** with memory-efficient incremental indexing, maintaining real-time consistency across millions of product documents; built deduplicating ingestion pipelines to pre-populate RAG retrieval indices (**¡1s index refresh, 99.9% consistency**).

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Managed distributed ML data pipelines on Kubernetes (Ray, MLflow); built FastAPI model serving APIs with TensorFlow Serving; automated CI/CD with Terraform, Docker, and ArgoCD; provisioned IaC and managed horizontal data partitioning and lineage tracking across ephemeral storage (**15+ models, 99.9% deploy success rate**).

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (Remote RA, Feb 2024 – Nov 2025) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.